第五章　偏微分

5.0 微積分函數分類

1. 單變數函數：$\mathbb{R} \to \mathbb{R}$
2. 向量值函數：$\mathbb{R} \to \mathbb{R}^n$，$n > 1$ (5.2)
3. 多變數函數：$\mathbb{R}^n \to \mathbb{R}$，$n > 1$ (5.3)
4. 多變數向量值函數：$\mathbb{R}^n \to \mathbb{R}^m$，$m$、$n > 1$

5.1 空間向量

定義 (符號)**.**

- 向量：$\mathbf{a}$，$\mathbf{x}$
- 分量形式：$\mathbf{a} = \langle a_1, a_2, a_3 \rangle$，$a_1$、$a_2$、$a_3 \in \mathbb{R}$；$\mathbf{a} \in \mathbb{R}^3$。
- 向量長度：$|\mathbf{a}| = |\langle a_1, a_2, a_3 \rangle| = \sqrt{a_1^2 + a_2^2 + a_3^2}$。
- 三維直角座標系單位向量：$\widehat{\boldsymbol{\imath}} = \langle 1, 0, 0 \rangle$，$\widehat{\boldsymbol{\jmath}} = \langle 0, 1, 0 \rangle$，$\widehat{\mathbf{k}} = \langle 0, 0, 1 \rangle$
- $\mathbf{a} = \langle a_1, a_2, a_3 \rangle \equiv a_1 \widehat{\boldsymbol{\imath}} + a_2 \widehat{\boldsymbol{\jmath}} + a_3 \widehat{\mathbf{k}}$

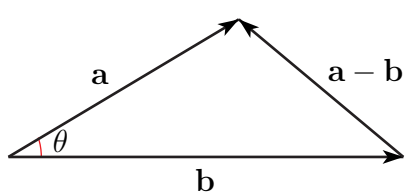
定義 (內積 / 點積)**.** 給定 n ($n \geqslant 2$) 維向量 $\mathbf{a} = \langle a_1, a_2, \ldots, a_n \rangle$，$\mathbf{b} = \langle b_1, b_2, \ldots, b_n \rangle$，則 $\mathbf{a}$ 與 $\mathbf{b}$ 的內積 / 點積 (inner product / dot product) 定義為 $\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{n} a_i b_i$。

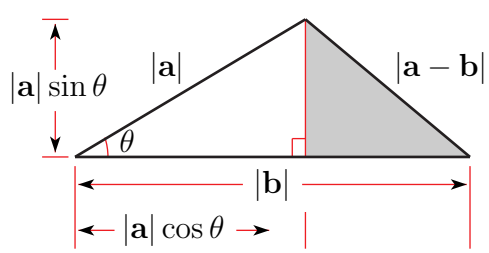
性質. 給定 $\mathbf{a}$，$\mathbf{b}$，$\mathbf{c} \in \mathbb{R}^n$，$n \geqslant 2$，$s \in \mathbb{R}$。則

- $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$，$0 \leqslant \theta \leqslant \pi$ 為 $\mathbf{a}$ 與 $\mathbf{b}$ 的夾角。

且

1. $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$
2. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
3. $(s\mathbf{a}) \cdot \mathbf{b} = s(\mathbf{a} \cdot \mathbf{b})$
4. $\mathbf{0} \cdot a = 0$
5. $\mathbf{a} \cdot \mathbf{b} = 0 \iff \mathbf{a} = \mathbf{0} \lor \mathbf{b} = \mathbf{0} \lor \mathbf{a} \perp \mathbf{b}$
6. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$，$(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c}$

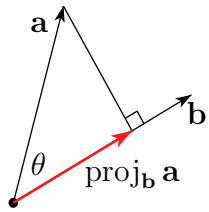


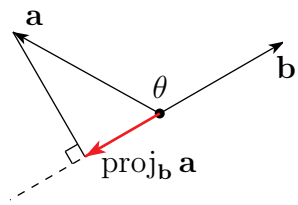


證. 由上圖，$|\mathbf{a} - \mathbf{b}|^2 = \big(|\mathbf{b}| - |\mathbf{a}| \cos\theta\big)^2 + \big(|\mathbf{a}| \sin\theta\big)^2 = |\mathbf{b}|^2 - 2\,|\mathbf{a}|\,|\mathbf{b}| \cos\theta + |\mathbf{a}|^2 \cos^2\theta + |\mathbf{a}|^2 \sin^2\theta = |\mathbf{b}|^2 - 2\,|\mathbf{a}|\,|\mathbf{b}| \cos\theta + |\mathbf{a}|^2$，又 $|\mathbf{a} - \mathbf{b}|^2 = (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) = \mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{a} - \mathbf{b} \cdot \mathbf{b} = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2\,\mathbf{a} \cdot \mathbf{b} = |\mathbf{b}|^2 - 2\,|\mathbf{a}|\,|\mathbf{b}| \cos\theta + |\mathbf{a}|^2$，故 $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$。

定義 (投影)**.** 給定向量 $\mathbf{a}$，$\mathbf{b}$，則 $\mathbf{a}$ 在 $\mathbf{b}$ 方向上的投影 (projection) $\operatorname{proj}_{\mathbf{b}} \mathbf{a}$ 定義為 $\operatorname{proj}_{\mathbf{b}} \mathbf{a} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b}$。

定義 (外積 / 叉積)**.** 給定三維向量 $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$，$\mathbf{b} = \langle b_1, b_2, b_3 \rangle$，則 $\mathbf{a}$ 與 $\mathbf{b}$ 的外積 / 叉積 (outer product / cross product) 定義為 $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \widehat{\boldsymbol{\imath}} & \widehat{\boldsymbol{\jmath}} & \widehat{\mathbf{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle$。



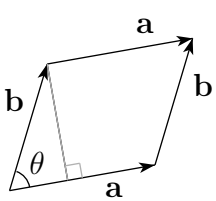


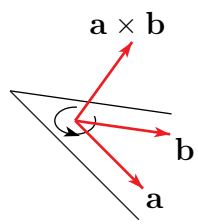
性質. $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c} \in \mathbb{R}^3$, $s \in \mathbb{R}$, 則

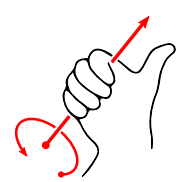
- $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}|\,\sin\theta$, $0 \leqslant \theta \leqslant \pi$ 為 $\mathbf{a}$ 與 $\mathbf{b}$ 的夾角; $|\mathbf{a} \times \mathbf{b}|$ 為 $\mathbf{a}$ 與 $\mathbf{b}$ 張成之平行四邊形面積。
- $\mathbf{a} \times \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}|\,\sin\theta\,\widehat{\mathbf{n}}$, $0 \leqslant \theta \leqslant \pi$ 為 $\mathbf{a}$ 與 $\mathbf{b}$ 的夾角, $(\mathbf{a}, \mathbf{b}, \widehat{\mathbf{n}})$ 滿足右手定則且 $|\widehat{\mathbf{n}}| = 1$, $\widehat{\mathbf{n}} \perp \mathbf{a}$, $\widehat{\mathbf{n}} \perp \mathbf{b}$。
- $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$ 為 $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ 張成之平行六面體體積。

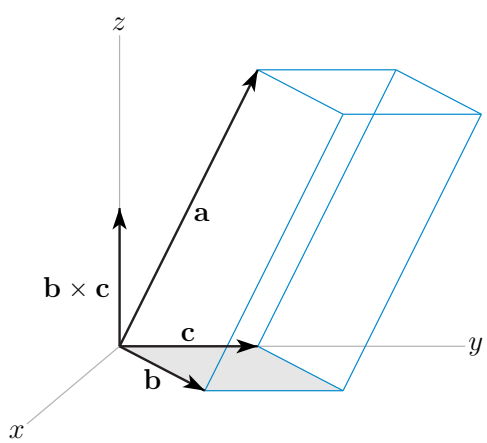
且

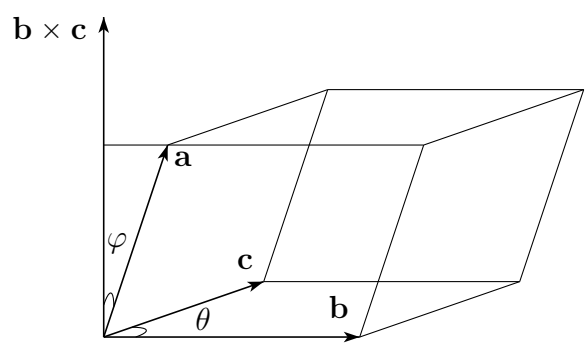
1. $\mathbf{a} \times \mathbf{b} \perp \mathbf{a}$, $\mathbf{a} \times \mathbf{b} \perp \mathbf{b}$
2. $\widehat{\imath} \times \widehat{\jmath} = \widehat{\mathbf{k}}$, $\widehat{\jmath} \times \widehat{\mathbf{k}} = \widehat{\imath}$, $\widehat{\mathbf{k}} \times \widehat{\imath} = \widehat{\jmath}$
3. $\mathbf{a} \times \mathbf{b} = 0 \iff \mathbf{a} = \mathbf{0} \lor \mathbf{b} = \mathbf{0} \lor \mathbf{a} \parallel \mathbf{b}$
4. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
5. $(s\mathbf{a}) \times \mathbf{b} = \mathbf{a} \times (s\mathbf{b}) = s(\mathbf{a} \times \mathbf{b})$
6. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
7. $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$
8. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$
9. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}\,(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}\,(\mathbf{a} \cdot \mathbf{b})$ (baccab 規則)











證.

- $|\mathbf{a}\times\mathbf{b}|^2 = (\mathbf{a}\times\mathbf{b})\cdot(\mathbf{a}\times\mathbf{b}) = (a_2b_3-a_3b_2)^2+(a_3b_1-a_1b_3)^2+(a_1b_2-a_2b_1)^2 = a_2^2b_3^2-2a_2b_3a_3b_2+a_3^2b_2^2+a_3^2b_1^2-2a_3b_1a_1b_3+a_1^2b_3^2+a_1^2b_2^2-2a_1b_2a_2b_1+a_2^2b_1^2$, 而 $|\mathbf{a}|^2\,|\mathbf{b}|^2\sin^2\theta = |\mathbf{a}|^2\,|\mathbf{b}|^2(1-\cos^2\theta) = |\mathbf{a}|^2\,|\mathbf{b}|^2 - (\mathbf{a}\cdot\mathbf{b})^2 = \left(a_1^2+a_2^2+a_3^2\right)\left(b_1^2+b_2^2+b_3^2\right) - \left(a_1b_1+a_2b_2+a_3b_3\right)^2 = a_1^2b_2^2+a_1^2b_3^2+a_2^2b_1^2+a_2^2b_3^2+a_3^2b_1^2+a_3^2b_2^2-\left(2a_1b_1a_2b_2+2a_1b_1a_3b_3+2a_2b_2a_3b_3\right)$, 故 $|\mathbf{a}\times\mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}|\sin\theta$ 。

- $\mathbf{a}\cdot(\mathbf{a}\times\mathbf{b}) = a_1(a_2b_3-a_3b_2)+a_2(a_3b_1-a_1b_3)+a_3(a_1b_2-a_2b_1) = 0$,
  $\mathbf{b}\cdot(\mathbf{a}\times\mathbf{b}) = b_1(a_2b_3-a_3b_2)+b_2(a_3b_1-a_1b_3)+b_3(a_1b_2-a_2b_1) = 0$

- $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \langle a_1,a_2,a_3\rangle\cdot\langle b_2c_3-b_3c_2\,,\,b_3c_1-b_1c_3\,,\,b_1c_2-b_2c_1\rangle = a_1b_2c_3-a_1b_3c_2+a_2b_3c_1-a_2b_1c_3+a_3b_1c_2-a_3b_2c_1$，又 $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c} = \langle a_2b_3-a_3b_2\,,\,a_3b_1-a_1b_3\,,\,a_1b_2-a_2b_1\rangle\cdot\langle c_1,c_2,c_3\rangle = a_2b_3c_1-a_3b_2c_1+a_3b_1c_2-a_1b_3c_2+a_1b_2c_3-a_2b_1c_3$。
  另證：$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \langle a_1,a_2,a_3\rangle\cdot\det\begin{vmatrix}\widehat{\imath} & \widehat{\jmath} & \widehat{\mathbf{k}}\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix} = a_1\det\begin{vmatrix}b_2 & b_3\\ c_2 & c_3\end{vmatrix} - a_2\det\begin{vmatrix}b_1 & b_3\\ c_1 & c_3\end{vmatrix} + a_3\det\begin{vmatrix}b_1 & b_2\\ c_1 & c_2\end{vmatrix}$
  $= \det\begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}$，而 $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c} = \det\begin{vmatrix}\widehat{\imath} & \widehat{\jmath} & \widehat{\mathbf{k}}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}\cdot\langle c_1,c_2,c_3\rangle = c_1\det\begin{vmatrix}a_2 & a_3\\ b_2 & b_3\end{vmatrix} - c_2\det\begin{vmatrix}a_1 & a_3\\ b_1 & b_3\end{vmatrix} + c_3\det\begin{vmatrix}a_1 & a_2\\ b_1 & b_2\end{vmatrix} = \det\begin{vmatrix}c_1 & c_2 & c_3\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}$，故 $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$。

- $\mathbf{b}\times\mathbf{c} = (b_2c_3-b_3c_2)\widehat{\imath} - (b_1c_3-b_3c_1)\widehat{\jmath} + (b_1c_2-b_2c_1)\widehat{\mathbf{k}}$，故 $\mathbf{a}\times(\mathbf{b}\times\mathbf{c})$
  $= \det\begin{vmatrix}\widehat{\imath} & \widehat{\jmath} & \widehat{\mathbf{k}}\\ a_1 & a_2 & a_3\\ b_2c_3-b_3c_2 & -b_1c_3+b_3c_1 & b_1c_2-b_2c_1\end{vmatrix} = \widehat{\imath}\big(a_2(b_1c_2-b_2c_1)-a_3(-b_1c_3+b_3c_1)\big) - \widehat{\jmath}\big(a_1(b_1c_2-b_2c_1)-a_3(b_2c_3-b_3c_2)\big) + \widehat{\mathbf{k}}\big(a_1(-b_1c_3+b_3c_1)-a_2(b_2c_3-b_3c_2)\big)$。而 $\mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{c}(\mathbf{a}\cdot\mathbf{b}) = (a_1c_1+a_2c_2+a_3c_3)(b_1\widehat{\imath}+b_2\widehat{\jmath}+b_3\widehat{\mathbf{k}}) - (a_1b_1+a_2b_2+a_3b_3)(c_1\widehat{\imath}+c_2\widehat{\jmath}+c_3\widehat{\mathbf{k}}) = \widehat{\imath}\big(a_1b_1c_1+a_2b_1c_2+a_3b_1c_3-a_1b_1c_1-a_2b_2c_1-a_3b_3c_1\big) + \widehat{\jmath}\big(a_1b_2c_1+a_2b_2c_2+a_3b_2c_3-a_1b_1c_2-a_2b_2c_2-a_3b_3c_2\big) + \widehat{\mathbf{k}}\big(a_1b_3c_1+a_2b_3c_2+a_3b_3c_3-a_1b_1c_3-a_2b_2c_3-a_3b_3c_3\big) = \widehat{\imath}(a_2b_1c_2+a_3b_1c_3-a_2b_2c_1-a_3b_3c_1)+\widehat{\jmath}(a_1b_2c_1+a_3b_2c_3-a_1b_1c_2-a_3b_3c_2)+\widehat{\mathbf{k}}(a_1b_3c_1+a_2b_3c_2-a_1b_1c_3-a_2b_2c_3)$，故 $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{c}(\mathbf{a}\cdot\mathbf{b})$。

例. 若 $\mathbf{a}$，$\mathbf{b}$，$\mathbf{c}$，$\mathbf{d}\in\mathbb{R}^3$，則

1. $\mathbf{a}\times(\mathbf{b}\times\mathbf{c})+\mathbf{b}\times(\mathbf{c}\times\mathbf{a})+\mathbf{c}\times(\mathbf{a}\times\mathbf{b}) = 0$
2. $(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d}) = (\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d})-(\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})$
3. $(\mathbf{a}\times\mathbf{b})\cdot((\mathbf{b}\times\mathbf{c})\times(\mathbf{c}\times\mathbf{a})) = (\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}))^2$

解.

1. $\mathbf{a}\times(\mathbf{b}\times\mathbf{c})+\mathbf{b}\times(\mathbf{c}\times\mathbf{a})+\mathbf{c}\times(\mathbf{a}\times\mathbf{b}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c})-\mathbf{c}(\mathbf{a}\cdot\mathbf{b})+\mathbf{c}(\mathbf{b}\cdot\mathbf{a})-\mathbf{a}(\mathbf{b}\cdot\mathbf{c})+\mathbf{a}(\mathbf{c}\cdot\mathbf{b})-\mathbf{b}(\mathbf{c}\cdot\mathbf{a}) = 0$
2. $(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d}) = \mathbf{c}\cdot(\mathbf{d}\times(\mathbf{a}\times\mathbf{b})) = \mathbf{c}\cdot(\mathbf{a}(\mathbf{d}\cdot\mathbf{b})-\mathbf{b}(\mathbf{d}\cdot\mathbf{a})) = (\mathbf{c}\cdot\mathbf{a})(\mathbf{d}\cdot\mathbf{b})-(\mathbf{c}\cdot\mathbf{b})(\mathbf{d}\cdot\mathbf{a}) = (\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d})-(\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})$
3. $(\mathbf{a}\times\mathbf{b})\cdot((\mathbf{b}\times\mathbf{c})\times(\mathbf{c}\times\mathbf{a})) = (\mathbf{b}\times\mathbf{c})\cdot((\mathbf{c}\times\mathbf{a})\times(\mathbf{a}\times\mathbf{b})) = (\mathbf{b}\times\mathbf{c})\cdot(\mathbf{a}\,((\mathbf{c}\times\mathbf{a})\cdot\mathbf{b})-\mathbf{b}\,((\mathbf{c}\times\mathbf{a})\cdot\mathbf{a})) = (\mathbf{b}\times\mathbf{c})\cdot(\mathbf{a}\,((\mathbf{c}\times\mathbf{a})\cdot\mathbf{b})) = (\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}))^2$

性質 (常用公式).

- 點 $p=(p_1,\,p_2,\,p_3)$ 與平面 $ax+by+cz+d=0$ 距離為 $\dfrac{|ap_1+bp_2+cp_3+d|}{\sqrt{a^2+b^2+c^2}}$。
- 若 $\mathbf{a}=\langle a_1,\,a_2,\,a_3\rangle$，$\mathbf{b}=\langle b_1,\,b_2,\,b_3\rangle$，$\mathbf{c}=\langle c_1,\,c_2,\,c_3\rangle$，$\mathbf{d}=\langle d_1,\,d_2,\,d_3\rangle$，三維空間中兩直線 $\langle a_1+b_1\,s,\,a_2+b_2\,s,\,a_3+b_3\,s\rangle$，$\langle c_1+d_1\,t,\,c_2+d_2\,t,\,c_3+d_3\,t\rangle$，$s$，$t\in\mathbb{R}$ 之距離為 $\dfrac{|(\mathbf{a}-\mathbf{c})\cdot(\mathbf{b}\times\mathbf{d})|}{|\mathbf{b}\times\mathbf{d}|}$。

解.

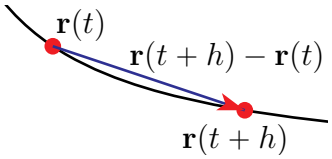
- 平面 S：$ax+by+cz+d=0$ 法向量為 $\mathbf{n}=\langle a,\,b,\,c\rangle$；點 $p=(p_1,\,p_2,\,p_3)$ 與投影至平面 S 之點 $o=(x,\,y,\,z)$ 所形成之向量平行於 $\mathbf{n}$，故 $(x,\,y,\,z)=(p_1+a\,t,\,p_2+b\,t,\,p_3+c\,t)$，$t\in\mathbb{R}$ 為待定常數。又 o 位於平面 S 上，故 $a(p_1+a\,t)+b(p_2+b\,t)+c(p_3+c\,t)+d=0 \implies t=\dfrac{-(ap_1+bp_2+cp_3+d)}{a^2+b^2+c^2}$，所求距離 $\overline{op}=|\langle a\,t,\,b\,t,\,c\,t\rangle| = \sqrt{a^2+b^2+c^2}\,|t| = \dfrac{|ap_1+bp_2+cp_3+d|}{\sqrt{a^2+b^2+c^2}}$。

- 兩直線點分別為 $\mathbf{a}$、$\mathbf{c}$, 方向向量分別為 $\mathbf{b}$、$\mathbf{d}$; $\mathbf{b}\times\mathbf{d}$ 同時垂直於兩直線, 所求距離即為 $|\mathrm{proj}_{\mathbf{b}\times\mathbf{d}}(\mathbf{a}-\mathbf{c})| = \dfrac{|(\mathbf{a}-\mathbf{c})\cdot(\mathbf{b}\times\mathbf{d})|}{|\mathbf{b}\times\mathbf{d}|}$。

5.2 向量值函數

定義. 向量值函數 $\mathbf{r}(t):\mathbb{R}\to\mathbb{R}^n$, $n>1$, 其微分為

$$\mathbf{r}'(t) = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}(t) = \lim_{h\to 0}\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}$$



若 $\mathbf{r}(t) = x(t)\widehat{\boldsymbol{\imath}} + y(t)\widehat{\boldsymbol{\jmath}} + z(t)\widehat{\mathbf{k}}$, 則 $\mathbf{r}'(t) = x'(t)\widehat{\boldsymbol{\imath}} + y'(t)\widehat{\boldsymbol{\jmath}} + z'(t)\widehat{\mathbf{k}}$。

註. 向量值函數多用在空間曲線表示式中: (曲線) $\equiv$ (位置向量)。

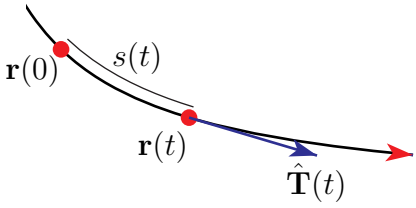
定理 (微分規則)**.** 令 $\mathbf{a}(t)$、$\mathbf{b}(t)$ 為 $t\in\mathbb{R}$ 可微 $\mathbb{R}^n$ 向量值函數, α、$\beta\in\mathbb{R}$, $\gamma(t)$、$s(t)$ 為 $t\in\mathbb{R}$ 可微實函數, 則

1. (線性) $\dfrac{\mathrm{d}}{\mathrm{d}t}\big(\alpha\,\mathbf{a}(t)+\beta\,\mathbf{b}(t)\big) = \alpha\,\mathbf{a}'(t)+\beta\,\mathbf{b}'(t)$
2. (乘積) $\dfrac{\mathrm{d}}{\mathrm{d}t}\big(\gamma(t)\mathbf{b}(t)\big) = \gamma'(t)\mathbf{b}(t)+\gamma(t)\mathbf{b}'(t)$
3. (內積) $\dfrac{\mathrm{d}}{\mathrm{d}t}\big(\mathbf{a}(t)\cdot\mathbf{b}(t)\big) = \mathbf{a}'(t)\cdot\mathbf{b}(t)+\mathbf{a}(t)\cdot\mathbf{b}'(t)$
4. (外積) $\dfrac{\mathrm{d}}{\mathrm{d}t}\big(\mathbf{a}(t)\times\mathbf{b}(t)\big) = \mathbf{a}'(t)\times\mathbf{b}(t)+\mathbf{a}(t)\times\mathbf{b}'(t)$
5. (合成) $\dfrac{\mathrm{d}}{\mathrm{d}t}\big(\mathbf{a}\big(s(t)\big)\big) = \mathbf{a}'\big(s(t)\big)\,s'(t)$

性質. 給定曲線 $\mathbf{r}(t)$。

1. 令 $\widehat{\mathbf{T}}(t)$ 為曲線在點 $\mathbf{r}(t)$ 並指向 t 遞增方向之單位切線向量, 則 $\widehat{\mathbf{T}}(t) = \dfrac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$, $\mathbf{r}'(t)\neq\mathbf{0}$。
2. 令 $s(t)$ 為曲線介於點 $\mathbf{r}(0)$ 與 $\mathbf{r}(t)$ 間之弧長, 則

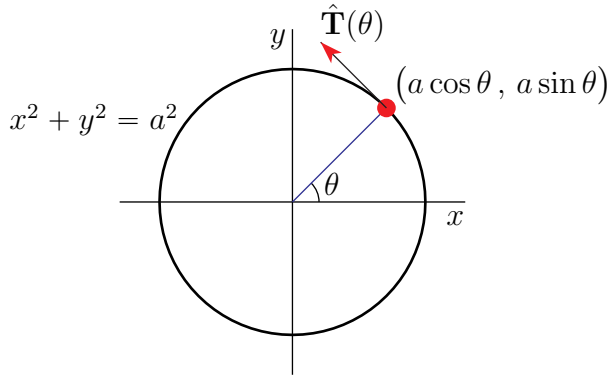
$$\frac{\mathrm{d}s}{\mathrm{d}t}(t) = \left|\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}(t)\right|$$
$$s(T)-s(T_0) = \int_{T_0}^{T}\left|\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}(t)\right|\,\mathrm{d}t$$

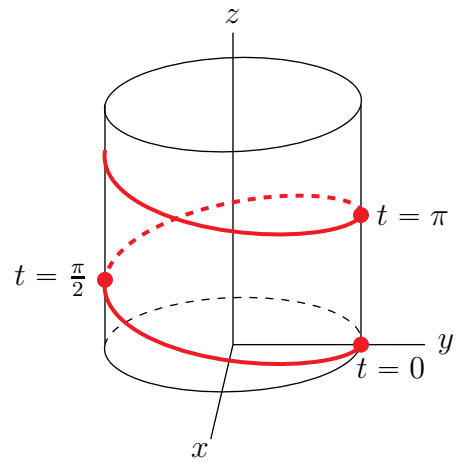


3. 若以弧長為參數, 亦即 $t=s$ 使得 $\dfrac{\mathrm{d}t}{\mathrm{d}s} = \dfrac{\mathrm{d}s}{\mathrm{d}s} = 1$, 則 $\left|\dfrac{\mathrm{d}\mathbf{r}}{\mathrm{d}s}(s)\right| = 1$, $\widehat{\mathbf{T}}(s) = \mathbf{r}'(s)$。

性質. 給定位置向量 $\mathbf{r}(t) = \langle x(t)\,,\,y(t)\,,\,z(t)\rangle$, 則時點 t 之

- 速度 $\mathbf{v}(t) = \mathbf{r}'(t) = x'(t)\widehat{\boldsymbol{\imath}} + y'(t)\widehat{\boldsymbol{\jmath}} + z'(t)\widehat{\mathbf{k}}$
- 速率 $\dfrac{\mathrm{d}s}{\mathrm{d}t}(t) = |\mathbf{v}(t)| = |\mathbf{r}'(t)| = \sqrt{(x'(t)^2+y'(t)^2+z'(t)^2}$
- 加速度 $\mathbf{a}(t) = \mathbf{r}''(t) = \mathbf{v}'(t) = x''(t)\widehat{\boldsymbol{\imath}} + y''(t)\widehat{\boldsymbol{\jmath}} + z''(t)\widehat{\mathbf{k}}$





時點 T_0 與 T 間經過距離為 $s(T)-s(T_0)=\displaystyle\int_{T_0}^{T}\left|\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}(t)\right|\mathrm{d}t=\int_{T_0}^{T}\sqrt{(x'(t)^2+y'(t)^2+z'(t)^2}\,\mathrm{d}t$

例. 圓 $x^2+y^2=a^2$ 之曲線表示式為 $\mathbf{r}(\theta)=\langle a\cos\theta,\,a\sin\theta\rangle$，$0\leqslant\theta\leqslant 2\pi$。$\mathbf{r}'(\theta)=\langle -a\sin\theta,\,a\cos\theta\rangle$，$\widehat{\mathbf{T}}(\theta)=\dfrac{\mathbf{r}'(\theta)}{|\mathbf{r}'(\theta)|}=\langle -\sin\theta,\,\cos\theta\rangle$，$\dfrac{\mathrm{d}s}{\mathrm{d}\theta}(\theta)=\left|\mathbf{r}'(\theta)\right|=a$，$s(\Theta)-s(0)=\displaystyle\int_0^{\Theta}\left|\mathbf{r}'(\theta)\right|\mathrm{d}\theta=a\Theta$。

例 (螺旋線弧長)**.** 求 $\mathbf{r}(t)=6\sin 2t\,\widehat{\boldsymbol{\imath}}+6\cos 2t\,\widehat{\boldsymbol{\jmath}}+5t\,\widehat{\mathbf{k}}$ 介於 $t=0$ 與 $t=\pi$ 間弧長。

解. $\mathbf{r}(t)=6\sin 2t\,\widehat{\boldsymbol{\imath}}+6\cos 2t\,\widehat{\boldsymbol{\jmath}}+5t\,\widehat{\mathbf{k}}\implies\mathbf{r}'(t)=12\cos 2t\,\widehat{\boldsymbol{\imath}}-12\sin 2t\,\widehat{\boldsymbol{\jmath}}+5\,\widehat{\mathbf{k}}$。則 $\dfrac{\mathrm{d}s}{\mathrm{d}t}(t)=\left|\mathbf{r}'(t)\right|=\sqrt{12^2\cos^2 2t+12^2\sin^2 2t+5^2}=\sqrt{12^2+5^2}=13$，$\widehat{\mathbf{T}}(t)=\dfrac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}=\dfrac{12}{13}\cos 2t\,\widehat{\boldsymbol{\imath}}-\dfrac{12}{13}\sin 2t\,\widehat{\boldsymbol{\jmath}}+\dfrac{5}{13}\,\widehat{\mathbf{k}}$，
$s(\pi)-s(0)=\displaystyle\int_0^{\pi}\left|\mathbf{r}'(t)\right|\mathrm{d}t=13\pi$。

例 (螺旋線弧長)**.** 求 $\mathbf{r}(t)=6\sin 2t\,\widehat{\boldsymbol{\imath}}+6\cos 2t\,\widehat{\boldsymbol{\jmath}}+5t\,\widehat{\mathbf{k}}$ 介於 $t=0$ 與 $t=\pi$ 間弧長。

解. $\mathbf{r}(t)=6\sin 2t\,\widehat{\boldsymbol{\imath}}+6\cos 2t\,\widehat{\boldsymbol{\jmath}}+5t\,\widehat{\mathbf{k}}\implies\mathbf{r}'(t)=12\cos 2t\,\widehat{\boldsymbol{\imath}}-12\sin 2t\,\widehat{\boldsymbol{\jmath}}+5\,\widehat{\mathbf{k}}$。則 $\dfrac{\mathrm{d}s}{\mathrm{d}t}(t)=\left|\mathbf{r}'(t)\right|=\sqrt{12^2\cos^2 2t+12^2\sin^2 2t+5^2}=\sqrt{12^2+5^2}=13$，$\widehat{\mathbf{T}}(t)=\dfrac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}=\dfrac{12}{13}\cos 2t\,\widehat{\boldsymbol{\imath}}-\dfrac{12}{13}\sin 2t\,\widehat{\boldsymbol{\jmath}}+\dfrac{5}{13}\,\widehat{\mathbf{k}}$，
$s(\pi)-s(0)=\displaystyle\int_0^{\pi}\left|\mathbf{r}'(t)\right|\mathrm{d}t=13\pi$。

例. 求 $\mathbf{r}(t)=\left\langle e^{3t},\,e^{-3t},\,3\sqrt{2}\,t\right\rangle$ 介於 $t=0$ 與 $t=\dfrac{1}{3}$ 間弧長。

解. $\mathbf{r}'(t)=\left\langle 3e^{3t},\,-3e^{-3t},\,3\sqrt{2}\right\rangle$,$s\left(\dfrac{1}{3}\right)-s(0)=\displaystyle\int_0^{\frac{1}{3}}\left|\mathbf{r}'(t)\right|\mathrm{d}t=\int_0^{\frac{1}{3}}\sqrt{9e^{6t}+9e^{-6t}+18}\,\mathrm{d}t=3\int_0^{\frac{1}{3}}\sqrt{e^{6t}+e^{-6t}+2}\,\mathrm{d}t$
$=3\displaystyle\int_0^{\frac{1}{3}}\sqrt{(e^{3t}+e^{-3t})^2}\,\mathrm{d}t=3\int_0^{\frac{1}{3}}(e^{3t}+e^{-3t})\,\mathrm{d}t=e^{3t}-e^{-3t}\Big|_0^{\frac{1}{3}}=e-\dfrac{1}{e}$。

例. 求 $\mathbf{r}(t)=\langle t,\,2t,\,t^2\rangle$ 介於 $t=1$ 與 $t=3$ 間弧長。

解. $\mathbf{r}'(t)=\langle 1,\,2,\,2t\rangle$，$s(3)-s(1)=\displaystyle\int_1^3\left|\mathbf{r}'(t)\right|\mathrm{d}t=\int_1^3\sqrt{5+4t^2}\,\mathrm{d}t=\dfrac{6\sqrt{41}-6-5\ln 5+5\ln(\sqrt{41}+6)}{4}$。
利用 $\displaystyle\int\sqrt{x^2+a^2}\,\mathrm{d}x=\dfrac{x\sqrt{x^2+a^2}+a^2\ln\left|\sqrt{x^2+a^2}+x\right|}{2}+c$。

5.3 極限與微分

定義 (多變數函數)**.** 令 $U \subseteq \mathbb{R}^n$，$n > 1$，從 $U \to \mathbb{R}$ 的映射 $f(x_1, x_2, \ldots, x_n) : U \to \mathbb{R}$ 稱為 U 上的 n 變數函數 (real-valued function of n variables)，其中 U 為定義域，$f(U)$ 為值域。

註. 若 $f(x_1, x_2, \ldots, x_n)$ 為 n 變數函數，可將 f 視為

- n 個實變數 $x_1, x_2, \ldots, x_n$ 的函數
- 向量 $\mathbf{x} = \langle x_1, x_2, \ldots, x_n \rangle$ 的函數
- $\mathbb{R}^n$ 中之點 $(x_1, x_2, \ldots, x_n)$ 的函數

定義 (圖形、等值曲線)**.** 令 $f(x, y)$ 為定義在 U 上的雙變數函數。

- 集合 $\left\{(x, y, z) \in \mathbb{R}^3 \,|\, z = f(x, y),\ (x, y) \in U\right\}$ 稱為 f 的圖形 (graph)。
- 給定常數 $k \in \mathbb{R}$，曲線 $f(x, y) = k$ 稱為 f 的等值曲線 (level / contour curve)。

若 $w = f(x, y, z)$ 為三變數函數，$f(x, y, z) = k$ 稱為 f 的等值曲面 (level surface)。

定義 (極限)**.** 令 f 為 n 變數函數。若對任意 $\varepsilon > 0$ 存在 $\delta > 0$ 使得對所有 $\mathbf{x} \in \operatorname{dom} f$ 滿足

$$0 < |\mathbf{x} - \mathbf{a}| < \delta \implies |f(\mathbf{x}) - L| < \varepsilon$$

則稱 f 在 $\mathbf{a}$ 的極限為 L，記為 $\lim\limits_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x}) = L$。

性質 (極限運算)**.** 令 $\mathbf{a} \in \mathbb{R}^n$，$c$、$F$、$G \in \mathbb{R}$，$D \subseteq \mathbb{R}^n$，$f, g : D \setminus \{\mathbf{a}\} \to \mathbb{R}$，$\gamma : \mathbb{R} \to \mathbb{R}$。若 $\lim\limits_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x}) = F$，$\lim\limits_{\mathbf{x} \to \mathbf{a}} g(\mathbf{x}) = G$，$\lim\limits_{t \to F} \gamma(t) = \gamma(F)$，則

1. $\lim\limits_{\mathbf{x} \to \mathbf{a}} \left[f(\mathbf{x}) \pm g(\mathbf{x})\right] = F \pm G$
2. $\lim\limits_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x})\, g(\mathbf{x}) = FG$
3. $\lim\limits_{\mathbf{x} \to \mathbf{a}} \dfrac{f(\mathbf{x})}{g(\mathbf{x})} = \dfrac{F}{G}$ 若 $G \neq \mathbf{0}$
4. $\lim\limits_{\mathbf{x} \to \mathbf{a}} \gamma\big(f(\mathbf{x})\big) = \gamma(F)$

例. 求 $\displaystyle\lim_{(x, y) \to (2, 3)} \frac{x + \sin y}{x^2 y^2 + 1}$。

解. $\displaystyle\lim_{(x, y) \to (2, 3)} \big(x + \sin y\big) = \lim_{(x, y) \to (2, 3)} x + \lim_{(x, y) \to (2, 3)} \sin y = \lim_{(x, y) \to (2, 3)} x + \sin\left(\lim_{(x, y) \to (2, 3)} y\right) = 2 + \sin 3$，
$\displaystyle\lim_{(x, y) \to (2, 3)} \big(x^2 y^2 + 1\big) = \lim_{(x, y) \to (2, 3)} x^2 y^2 + \lim_{(x, y) \to (2, 3)} 1 = \left(\lim_{(x, y) \to (2, 3)} x\right)\left(\lim_{(x, y) \to (2, 3)} x\right)\left(\lim_{(x, y) \to (2, 3)} y\right)\left(\lim_{(x, y) \to (2, 3)} y\right) +$
$1 = 2^2 3^2 + 1 = 37$，$\displaystyle\lim_{(x, y) \to (2, 3)} \frac{x + \sin y}{x^2 y^2 + 1} = \frac{\lim\limits_{(x, y) \to (2, 3)} (x + \sin y)}{\lim\limits_{(x, y) \to (2, 3)} (x^2 y^2 + 1)} = \frac{2 + \sin 3}{37}$

註.

- 單變數函數極限 $\lim\limits_{x \to a} f(x)$ 存在的充要條件為 $\lim\limits_{x \to a-} f(x)$ 與 $\lim\limits_{x \to a+} f(x)$ 均存在且相等。
- 多變數函數極限 $\lim\limits_{\mathbf{x} \to \mathbf{a}} f(\mathbf{x})$ 存在的充要條件為任意趨近 $\mathbf{a}$ 之路徑的極限均存在且相等。
- 求 $\lim\limits_{(x, y) \to (0, 0)} f(x, y)$ 常可將 (x, y) 轉成極座標 $x = r\cos\theta$，$y = r\sin\theta$ 後令 $r \to 0$ 觀察。

例. 求 $\displaystyle\lim_{(x, y) \to (0, 0)} \frac{x^2 y}{x^2 + y^2}$。

解. $\dfrac{x^2 y}{x^2 + y^2} = \dfrac{(r\cos\theta)^2 (r\sin\theta)}{r^2} = r\cos^2\theta\sin\theta$。由 $\left|r\cos^2\theta\sin\theta\right| \leqslant r \to 0$ 當 $r \to 0$，$\displaystyle\lim_{(x, y) \to (0, 0)} \frac{x^2 y}{x^2 + y^2} = 0$。

例. 求 $\displaystyle\lim_{(x, y) \to (0, 0)} \frac{x^2 - y^2}{x^2 + y^2}$。

解. 由 $f(x,y)=\dfrac{x^2-y^2}{x^2+y^2}=\dfrac{(r\cos\theta)^2-(r\sin\theta)^2}{r^2}=\cos^2\theta-\sin^2\theta=\cos(2\theta)$，$\displaystyle\lim_{(x,y)\to(0,0)}\frac{x^2-y^2}{x^2+y^2}=\text{DNE}$

例. 若 $f(x,y)=\begin{cases}\frac{(2x-y)^2}{x-y}, & x\neq y\\ 0, & x=y\end{cases}$，求 $\displaystyle\lim_{(x,y)\to(0,0)}f(x,y)$。

解.

- 延 $y=x-x^3$，$f(x,x-x^3)=\dfrac{(2x-x+x^3)^2}{x-x+x^3}=\dfrac{(x+x^3)^2}{x^3}=\dfrac{(1+x^2)^2}{x}\longrightarrow\begin{cases}+\infty, & x\to 0+\\ -\infty, & x\to 0-\end{cases}$
- 延 $y=x-ax^2$，$a\neq 0$：$\displaystyle\lim_{x\to 0}f(x,x-ax^2)=\lim_{x\to 0}\frac{(2x-x+ax^2)^2}{x-x+ax^2}=\lim_{x\to 0}\frac{(x+ax^2)^2}{ax^2}=\lim_{x\to 0}\frac{(1+ax)^2}{a}=\frac{1}{a}$
- 結論：$\displaystyle\lim_{(x,y)\to(0,0)}f(x,y)=\text{DNE}$

定義 (偏導函數、偏微分、偏導數).

- $f(x,y)$ 的 x-偏導函數定義為 $\dfrac{\partial f}{\partial x}(x,y)=\displaystyle\lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h}$；$f(x,y)$ 的 y-偏導函數定義為 $\dfrac{\partial f}{\partial y}(x,y)=\displaystyle\lim_{h\to 0}\frac{f(x,y+h)-f(x,y)}{h}$。
- 求 $f(x,y)$ 之 x-偏導函數之過程稱作「$f(x,y)$ 對 x 偏微分」。
- $f(x,y)$ 在 (a,b) 的 y-偏導數記為 $\dfrac{\partial f}{\partial y}(a,b)\equiv\left.\dfrac{\partial f}{\partial y}\right|_{(a,b)}$。

註.

- $\dfrac{\partial f}{\partial y}(x,y)$ 亦可記為 $\dfrac{\partial f}{\partial y}$、$f_y(x,y)$、$f_y$、$D_yf(x,y)$、$D_yf$、$D_2f(x,y)$、$D_2f$。
- 求 $\dfrac{\partial f}{\partial y}(x,y)$：將 $f(x,y)$ 中的 x 視作常數，然後對 y 微分。
- 求 $\dfrac{\partial f}{\partial y}(a,b)$：將 $f(x,y)$ 中的 x 視作常數，然後對 y 微分並代入 $x=a$、$y=b$。
- 以上符號 / 運算均可直接推廣至維度 >2 狀況。

例. $f(x,y)=x^3+y^2+4xy^2$，則 $\dfrac{\partial f}{\partial x}=\dfrac{\partial}{\partial x}(x^3)+\dfrac{\partial}{\partial x}(y^2)+\dfrac{\partial}{\partial x}(4xy^2)=3x^2+0+4y^2\dfrac{\partial}{\partial x}(x)=3x^2+4y^2$，$\dfrac{\partial f}{\partial y}=\dfrac{\partial}{\partial y}(x^3)+\dfrac{\partial}{\partial y}(y^2)+\dfrac{\partial}{\partial y}(4xy^2)=0+2y+4x\dfrac{\partial}{\partial y}(y^2)=2y+8xy$，$\dfrac{\partial f}{\partial x}(1,\,0)=3(1)^2+4(0)^2=3$，$\dfrac{\partial f}{\partial y}(1,\,0)=2(0)+8(1)(0)=0$。

例. $f(x,y)=y\cos x+xe^{xy}$，$\dfrac{\partial}{\partial x}e^{yx}=ye^{yx}$，$\dfrac{\partial f}{\partial x}(x,y)=y\dfrac{\partial}{\partial x}(\cos x)+e^{xy}\dfrac{\partial}{\partial x}(x)+x\dfrac{\partial}{\partial x}\left(e^{xy}\right)=-y\sin x+e^{xy}+xye^{xy}$，$\dfrac{\partial f}{\partial y}(x,y)=\cos x\dfrac{\partial}{\partial y}(y)+x\dfrac{\partial}{\partial y}\left(e^{xy}\right)=\cos x+x^2e^{xy}$

例. $f(x,y,z,t)=x\sin(y+2z)+t^2e^{3y}\ln z$，則 $\dfrac{\partial f}{\partial x}(x,y,z,t)=\sin(y+2z)$，$\dfrac{\partial f}{\partial y}(x,y,z,t)=x\cos(y+2z)+3t^2e^{3y}\ln z$，$\dfrac{\partial f}{\partial z}(x,y,z,t)=2x\cos(y+2z)+\dfrac{t^2e^{3y}}{z}$，$\dfrac{\partial f}{\partial t}(x,y,z,t)=2te^{3y}\ln z$。

例. 若 $f(x,y)=\begin{cases}\frac{\cos x-\cos y}{x-y} & x\neq y\\ 0 & x=y\end{cases}$

- $\forall x \neq y$，$f_x = \dfrac{-\sin x(x-y) - (\cos x - \cos y)}{(x-y)^2}$；無法由此求 $f_x(0,0)$。
- 由定義計算 $f_x(0,0)$:$f_x(0,0) = \lim\limits_{h\to 0} \dfrac{f(0+h,0)-f(0,0)}{h} = \lim\limits_{h\to 0} \dfrac{\frac{\cos h - 1}{h-0} - 0}{h} = \lim\limits_{h\to 0} \dfrac{\cos h - 1}{h^2} = \lim\limits_{h\to 0} \dfrac{-\sin h}{2h} = \lim\limits_{h\to 0} \dfrac{-\cos h}{2} = -\dfrac{1}{2}$。
- 由定義計算 $f_y(0,0)$:$f_y(0,0) = \lim\limits_{h\to 0} \dfrac{f(0,0+h)-f(0,0)}{h} = \lim\limits_{h\to 0} \dfrac{\frac{1-\cos h}{-h} - 0}{h} = \lim\limits_{h\to 0} \dfrac{\cos h - 1}{h^2} = \lim\limits_{h\to 0} \dfrac{-\sin h}{2h} = \lim\limits_{h\to 0} \dfrac{-\cos h}{2} = -\dfrac{1}{2}$。
- $\lim\limits_{(x,y)\to(0,0)} \dfrac{\cos x - \cos y}{x-y} = \lim\limits_{(x,y)\to(0,0)} \dfrac{-2\sin\frac{x+y}{2}\sin\frac{x-y}{2}}{x-y} = -\lim\limits_{(x,y)\to(0,0)} \sin\dfrac{x+y}{2} \lim\limits_{(x,y)\to(0,0)} \dfrac{\sin\frac{x-y}{2}}{\frac{x-y}{2}} = 0$，故 $f(x,y)$ 在 $(0,0)$ 連續。
- $f(x,y)$ 在 (a,a)，$a \neq 0$ 不連續：由定義 $\lim\limits_{(x,y)\to(a,a)} f(x,y) = \sin a$，但 $f(a,a) = 0$。

例. 若 x、y、z 滿足方程式 $z^5 + y^2 e^z + e^{2x} = 0$，求 $\dfrac{\partial z}{\partial x}(0,0)$。

解. 局域下 z 為 x、y 之函數；當 $x = y = 0$，原方程式為 $z(0,0)^5 = -1 \implies z(0,0) = -1$。令 $z \equiv z(x,y)$ 代入原方程式並對 x 偏微分得 $5\,z(x,y)^4 \dfrac{\partial z}{\partial x}(x,y) + y^2\, e^{z(x,y)} \dfrac{\partial z}{\partial x}(x,y) + 2\,e^{2x} = 0$；代入 $(x,\,y) = (0,\,0)$ 得 $5\,z(0,0)^4 \dfrac{\partial z}{\partial x}(0,0) + 2 = 0$，再由 $z(0,0) = -1$，$\dfrac{\partial z}{\partial x}(0,0) = -\dfrac{2}{5\,z(0,0)^4} = -\dfrac{2}{5}$。

例. 若 x、y、z 滿足方程式 $x^2 + y^2 + z^2 = 1$，證明 $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = z - \dfrac{1}{z}$。

解. 局域下 z 為 x、y 之函數；$x^2 + y^2 + z^2 = 1$ 對 x 偏微分得 $2x + 2z\dfrac{\partial z}{\partial x} = 0 \implies \dfrac{\partial z}{\partial x} = -\dfrac{x}{z}$；對 y 偏微分得 $2y + 2z\dfrac{\partial z}{\partial y} = 0 \implies \dfrac{\partial z}{\partial x} = -\dfrac{y}{z}$。故 $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = -\dfrac{x^2+y^2}{z} = \dfrac{z^2-1}{z} = z - \dfrac{1}{z}$。

例. 若 x、y、z 滿足方程式 $x\sin z - z^2 y = 1$，求 $\dfrac{\partial z}{\partial x}$，$\dfrac{\partial z}{\partial y}$。

解. 局域下 z 為 x、y 之函數；$x\sin z - z^2 y = 1$ 對 x 偏微分得 $\sin z + x\cos z\dfrac{\partial z}{\partial x} - 2yz\dfrac{\partial z}{\partial x} = 0 \implies \dfrac{\partial z}{\partial x} = \dfrac{x}{z}$；對 y 偏微分得 $2y + 2z\dfrac{\partial z}{\partial y} = 0 \implies \dfrac{\partial z}{\partial x} = -\dfrac{y}{z}$。故 $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = -\dfrac{x^2+y^2}{z} = \dfrac{z^2-1}{z} = z - \dfrac{1}{z}$。

定義 (高階偏導函數)**.** 給定可微雙變數函數 $f(x,y)$，

- $\dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial x}\right)(x,y) = \dfrac{\partial^2 f}{\partial x^2}(x,y) = f_{xx}(x,y)$
- $\dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial y}\right)(x,y) = \dfrac{\partial^2 f}{\partial x\partial y}(x,y) = f_{yx}(x,y)$
- $\dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial x}\right)(x,y) = \dfrac{\partial^2 f}{\partial y\partial x}(x,y) = f_{xy}(x,y)$
- $\dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial y}\right)(x,y) = \dfrac{\partial^2 f}{\partial y^2}(x,y) = f_{yy}(x,y)$

例. 令 $f(x,y) = e^{my}\cos(nx)$，則

- $f_x = -ne^{my}\sin(nx)$
- $f_{xx} = -n^2 e^{my}\cos(nx)$
- $f_{yx} = -mne^{my}\sin(nx)$
- $f_y = me^{my}\cos(nx)$
- $f_{yy} = m^2 e^{my}\cos(nx)$
- $f_{xy} = -mne^{my}\sin(nx)$

例. 令 $f(x,y) = e^{\alpha x + \beta y}$，則

- $f_x = \alpha e^{\alpha x+\beta y}$
- $f_y = \beta e^{\alpha x+\beta y}$
- $f_{xx} = \alpha^2 e^{\alpha x+\beta y}$
- $f_{yx} = \beta\alpha e^{\alpha x+\beta y}$
- $f_{xy} = \alpha\beta e^{\alpha x+\beta y}$
- $f_{yy} = \beta^2 e^{\alpha x+\beta y}$

對整數 $m, n \geqslant 0$，$\dfrac{\partial^{m+n} f}{\partial x^m \partial y^n} = \alpha^m \beta^n e^{\alpha x+\beta y}$。

例. 令 $f(x,y) = \ln(x^2+y^2)$，則

- $f_x = \dfrac{2x}{x^2+y^2}$
- $f_y = \dfrac{2y}{x^2+y^2}$
- $f_{xx} = \dfrac{(x^2+y^2)\cdot 2 - 2x\cdot 2x}{(x^2+y^2)^2} = \dfrac{2\,(y^2-x^2)}{(x^2+y^2)^2}$
- $f_{yy} = \dfrac{(x^2+y^2)\cdot 2 - 2y\cdot 2y}{(x^2+y^2)^2} = \dfrac{2\,(x^2-y^2)}{(x^2+y^2)^2}$

$f(x,y)$ 滿足 Laplace 方程式 $f_{xx} + f_{yy} = 0$。

例. 令 $f(x,y) = \tan^{-1}\dfrac{y}{x}$，則

- $f_x = \dfrac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\left(\dfrac{-y}{x^2}\right) = -\dfrac{y}{x^2+y^2}$
- $f_y = \dfrac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\left(\dfrac{1}{x}\right) = \dfrac{x}{x^2+y^2}$
- $f_{xx} = \dfrac{y\cdot 2x}{(x^2+y^2)^2} = \dfrac{2xy}{(x^2+y^2)^2}$
- $f_{yy} = -\dfrac{x\cdot 2y}{(x^2+y^2)^2} = -\dfrac{2xy}{(x^2+y^2)^2}$

$f(x,y)$ 滿足 Laplace 方程式 $f_{xx} + f_{yy} = 0$。

例. 若 $f(x_1,x_2,x_3,x_4) = x_1^4\,x_2^3\,x_3^2\,x_4$，則

- $\dfrac{\partial^4 f}{\partial x_1\partial x_2\partial x_3\partial x_4} = \dfrac{\partial^3}{\partial x_1\partial x_2\partial x_3}\left(x_1^4x_2^3x_3^2\right) = \dfrac{\partial^2}{\partial x_1\partial x_2}\left(2x_1^4x_2^3x_3\right) = \dfrac{\partial}{\partial x_1}\left(6x_1^4x_2^2x_3\right) = 24x_1^3x_2^2x_3$
- $\dfrac{\partial^4 f}{\partial x_4\partial x_3\partial x_2\partial x_1} = \dfrac{\partial^3}{\partial x_4\partial x_3\partial x_2}\left(4x_1^3x_2^3x_3^2x_4\right) = \dfrac{\partial^2}{\partial x_4\partial x_3}\left(12x_1^3x_2^2x_3^2x_4\right) = \dfrac{\partial}{\partial x_4}\left(24x_1^3x_2^2x_3x_4\right) = 24x_1^3x_2^2x_3$

定理 (Clairaut)**.** 若 $\dfrac{\partial^2 f}{\partial x\partial y}$ 與 $\dfrac{\partial^2 f}{\partial y\partial x}$ 均存在且在 $(x_0,\,y_0)$ 連續，則 $\dfrac{\partial^2 f}{\partial x\partial y}(x_0,y_0) = \dfrac{\partial^2 f}{\partial y\partial x}(x_0,y_0)$。

5.4 鏈鎖律

定理. 若 f 為 $x_1,\,x_2,\,\ldots,\,x_n$ 的可微函數，而 x_j 是 $t_1,\,t_2,\,\ldots,\,t_m$ 的可微函數，$n,\,m \geqslant 1$，則 f 為 $t_1,\,t_2,\,\ldots,\,t_m$ 的可微函數；輔助函數 $F(t_1,\,t_2,\,\ldots,\,t_m) \equiv f(x_1(t_1,\,t_2,\,\ldots,\,t_m),\,x_2(t_1,\,t_2,\,\ldots,\,t_m),\,\ldots,\,x_n(t_1,\,t_2,\,\ldots,\,t_m))$，則

$$\frac{\partial F}{\partial t_i} = \frac{\partial f}{\partial x_1}\,\frac{\partial x_1}{\partial t_i} + \frac{\partial f}{\partial x_2}\,\frac{\partial x_2}{\partial t_i} + \cdots + \frac{\partial f}{\partial x_n}\,\frac{\partial x_n}{\partial t_i}$$

例 $(n=m=2)$**.** 輔助函數 $F(s,t) \equiv f\big(x(s,t),\,y(s,t)\big)$，則

$$\begin{aligned}\frac{\partial F}{\partial s}(s,t) &= \frac{\partial f}{\partial x}\big(x(s,t),\,y(s,t)\big)\,\frac{\partial x}{\partial s}(s,t) + \frac{\partial f}{\partial y}\big(x(s,t),\,y(s,t)\big)\,\frac{\partial y}{\partial s}(s,t)\\ \frac{\partial F}{\partial t}(s,t) &= \frac{\partial f}{\partial x}\big(x(s,t),\,y(s,t)\big)\,\frac{\partial x}{\partial t}(s,t) + \frac{\partial f}{\partial y}\big(x(s,t),\,y(s,t)\big)\,\frac{\partial y}{\partial t}(s,t)\end{aligned}$$

例. 若 $f(x,y) = e^{xy}$，$x(s,t) = s$，$y(s,t) = \cos t$；$F(s,t) \equiv f\big(x(s,t),\,y(s,t)\big)$，求 $\dfrac{\partial F}{\partial s}$。

解.

- $\dfrac{\partial f}{\partial x} = y\,e^{xy} = y(s,t)\,e^{x(s,t)\,y(s,t)} = \cos t\,e^{s\cos t}$，$\dfrac{\partial f}{\partial y} = x\,e^{xy} = x(s,t)\,e^{x(s,t)\,y(s,t)} = s\,e^{s\cos t}$，$\dfrac{\partial x}{\partial s} = \dfrac{\partial s}{\partial s} = 1$，$\dfrac{\partial y}{\partial s} = \dfrac{\partial \cos t}{\partial s} = 0$，故 $\dfrac{\partial F}{\partial s} = \dfrac{\partial f}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial f}{\partial y}\dfrac{\partial y}{\partial s} = \cos t\,e^{s\cos t}\cdot 1 + s\;e^{s\cos t}\cdot 0 = \cos t\;e^{s\cos t}$。
- 直接寫出 $F(s,t)$ 並對 s 偏微分：$F(s,t) = f\big(x(s,t),\,y(s,t)\big) = e^{s\cdot\cos t}$，$\dfrac{\partial F}{\partial s} = e^{s\cdot\cos t}\,\cos t$。

例. 若 $f(x,y) = x^2 - y^2$，$x(t) = \cos t$，$y(t) = \sin t$，求 $\dfrac{\mathrm{d}f}{\mathrm{d}t}$。

解. 輔助函數 $F(t) \equiv f\big(x(t),\,y(t)\big)$，則

- $\dfrac{\partial f}{\partial x} = 2x = 2\cos t$，$\dfrac{\partial f}{\partial y} = -2y = -2\sin t$，$\dfrac{\mathrm{d}x}{\mathrm{d}t} = -\sin t$，$\dfrac{\mathrm{d}y}{\mathrm{d}t} = \cos t$，故 $\dfrac{\mathrm{d}F}{\mathrm{d}t} = \dfrac{\partial f}{\partial x}\dfrac{\mathrm{d}x}{\mathrm{d}t} + \dfrac{\partial f}{\partial y}\dfrac{\mathrm{d}y}{\mathrm{d}t} = (2\cos t)(-\sin t) + (-2\sin t)(\cos t) = -4\,\sin t\cos t$。
- 直接寫出 $F(t)$ 並對 t 微分：$F(t) = f\big(x(t),y(t)\big) = x(t)^2 - y(t)^2 = \cos^2 t - \sin^2 t$，故 $F'(t) = 2(\cos t)(-\sin t) - 2(\sin t)(\cos t) = -4\sin t\cos t$

例.

1. 令 $w = xy + z$，$x = \cos t$，$y = \sin t$，$z = t$，求 $\dfrac{\mathrm{d}w}{\mathrm{d}t}$ 與 $\dfrac{\mathrm{d}w}{\mathrm{d}t}\Big|_{t=0}$。
2. 令 $w = x + 2y + z^2$，$x = \dfrac{r}{s}$，$y = r^2 + \ln s$，$z = 2r$，求 $\dfrac{\partial w}{\partial r}$ 與 $\dfrac{\partial w}{\partial s}$。
3. 令 $w = x^4y + y^2z^3$，$x = rse^t$，$y = rs^2e^{-t}$，$z = r^2s\sin t$，求 $\dfrac{\partial w}{\partial s}\Big|_{(r,\,s,\,t)=(2,\,1,\,0)}$。

解.

1. $\dfrac{\partial w}{\partial x} = y = \sin t$，$\dfrac{\partial w}{\partial y} = x = \cos t$，$\dfrac{\partial w}{\partial z} = 1$，$\dfrac{\mathrm{d}x}{\mathrm{d}t} = -\sin t$，$\dfrac{\mathrm{d}y}{\mathrm{d}t} = \cos t$，$\dfrac{\mathrm{d}z}{\mathrm{d}t} = 1$，故 $\dfrac{\mathrm{d}w}{\mathrm{d}t} = \dfrac{\partial w}{\partial x}\dfrac{\mathrm{d}x}{\mathrm{d}t} + \dfrac{\partial w}{\partial y}\dfrac{\mathrm{d}y}{\mathrm{d}t} + \dfrac{\partial w}{\partial z}\dfrac{\mathrm{d}z}{\mathrm{d}t} = (\sin t)(-\sin t) + (\cos t)(\cos t) + (1)(1) = \cos 2t + 1$，$\dfrac{\mathrm{d}w}{\mathrm{d}t}\Big|_{t=0} = 1 + 1 = 2$。
2. $\dfrac{\partial w}{\partial x} = 1$，$\dfrac{\partial w}{\partial y} = 2$，$\dfrac{\partial w}{\partial z} = 2z = 4r$，$\dfrac{\partial x}{\partial r} = \dfrac{1}{s}$，$\dfrac{\partial y}{\partial r} = 2r$，$\dfrac{\partial z}{\partial r} = 2$，$\dfrac{\partial x}{\partial s} = -\dfrac{r}{s^2}$，$\dfrac{\partial y}{\partial s} = \dfrac{1}{s}$，$\dfrac{\partial z}{\partial s} = 0$，故

$$\begin{aligned}\frac{\partial w}{\partial r} &= \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r} = (1)\Big(\frac{1}{s}\Big) + (2)(2r) + (4r)(2) = \frac{1}{s} + 12r\\ \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s} = (1)\Big(-\frac{r}{s^2}\Big) + (2)\Big(\frac{1}{s}\Big) + (4r)(0) = -\frac{r}{s^2} + \frac{2}{s}\end{aligned}$$

3. $\dfrac{\partial w}{\partial x} = 4x^3y$, $\dfrac{\partial w}{\partial y} = x^4 + 2yz^3$, $\dfrac{\partial w}{\partial z} = 3y^2z^2$, $\dfrac{\partial x}{\partial s} = re^t$, $\dfrac{\partial y}{\partial s} = 2rse^{-t}$, $\dfrac{\partial z}{\partial s} = r^2\sin t$。當 $(r,\,s,\,t) = (2,\,1,\,0)$，$(x,\,y,\,z) = (2,\,2,\,0)$，故

$$\begin{aligned}\frac{\partial w}{\partial s}\Big|_{(r,\,s,\,t)=(2,\,1,\,0)} &= \Big(\frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s}\Big)\Big|_{(r,\,s,\,t)=(2,\,1,\,0)}\\ &= (4\cdot 2^3\cdot 2)(2) + (2^4 + 0)(2\cdot 2\cdot 1) + (0)(0) = 192\end{aligned}$$

例. 若 $z = f(x-y)$，證明 $\dfrac{\partial z}{\partial x} + \dfrac{\partial z}{\partial y} = 0$。

解. 令 $u = x - y$，則 $\dfrac{\partial z}{\partial x} = \dfrac{\mathrm{d}z}{\mathrm{d}u}\dfrac{\partial u}{\partial x} = \dfrac{\mathrm{d}z}{\mathrm{d}u}(1) = \dfrac{\mathrm{d}z}{\mathrm{d}u}$，$\dfrac{\partial z}{\partial y} = \dfrac{\mathrm{d}z}{\mathrm{d}u}\dfrac{\partial u}{\partial y} = \dfrac{\mathrm{d}z}{\mathrm{d}u}(-1) = -\dfrac{\mathrm{d}z}{\mathrm{d}u}$，故 $\dfrac{\partial z}{\partial x} + \dfrac{\partial z}{\partial y} = 0$。

例. 若 $z=f(x,y)$，$x=s+t$，$y=s-t$，證明 $\left(\frac{\partial z}{\partial x}\right)^2-\left(\frac{\partial z}{\partial y}\right)^2=\frac{\partial z}{\partial s}\frac{\partial z}{\partial t}$。

解. $\frac{\partial z}{\partial s}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial s}=\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}$，$\frac{\partial z}{\partial t}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial t}=\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}$，故 $\frac{\partial z}{\partial s}\frac{\partial z}{\partial t}=\left(\frac{\partial z}{\partial x}\right)^2-\left(\frac{\partial z}{\partial y}\right)^2$。

例. 若 $g(s,t)=f(s^2-t^2,t^2-s^2)$ 且 f 可微，證明 $t\frac{\partial g}{\partial s}+s\frac{\partial g}{\partial t}=0$。

解. 令 $u(s,t)=s^2-t^2$，$v(s,t)=t^2-s^2$，則 $g(s,t)=f(u(s,t),v(s,t))$。由鏈鎖律

$$\begin{aligned}\frac{\partial g}{\partial s}&=\frac{\partial f}{\partial u}\frac{\partial u}{\partial s}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial s}=\frac{\partial f}{\partial u}\cdot(2s)+\frac{\partial f}{\partial v}\cdot(-2s)\\ \frac{\partial g}{\partial t}&=\frac{\partial f}{\partial u}\frac{\partial u}{\partial t}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial t}=\frac{\partial f}{\partial u}\cdot(-2t)+\frac{\partial f}{\partial v}\cdot(2t)\end{aligned}$$

故 $t\frac{\partial g}{\partial s}+s\frac{\partial g}{\partial t}=t\left(\frac{\partial f}{\partial u}\cdot(2s)+\frac{\partial f}{\partial v}\cdot(-2s)\right)+s\left(\frac{\partial f}{\partial u}\cdot(-2t)+\frac{\partial f}{\partial v}\cdot(2t)\right)=0$。

例. 若 $u=f(x,y)$，$x=e^s\cos t$，$y=e^s\sin t$，證明 $\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=e^{-2s}\left(\left(\frac{\partial u}{\partial s}\right)^2+\left(\frac{\partial u}{\partial t}\right)^2\right)$。

解. 由鏈鎖律

$$\begin{aligned}\frac{\partial u}{\partial s}&=\frac{\partial u}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial s}=\frac{\partial u}{\partial x}\cdot(e^s\cos t)+\frac{\partial u}{\partial y}\cdot(e^s\sin t)\\ \frac{\partial u}{\partial t}&=\frac{\partial u}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial t}=\frac{\partial u}{\partial x}\cdot(-e^s\sin t)+\frac{\partial u}{\partial y}\cdot(e^s\cos t)\end{aligned}$$

故

$$\begin{aligned}\left(\frac{\partial u}{\partial s}\right)^2+\left(\frac{\partial u}{\partial t}\right)^2&=\left(\frac{\partial u}{\partial x}\cdot(e^s\cos t)+\frac{\partial u}{\partial y}\cdot(e^s\sin t)\right)^2+\left(\frac{\partial u}{\partial x}\cdot(-e^s\sin t)+\frac{\partial u}{\partial y}\cdot(e^s\cos t)\right)^2\\ &=\left(\frac{\partial u}{\partial x}\right)^2e^{2s}\cos^2 t+2\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}e^{2s}\cos t\sin t+\left(\frac{\partial u}{\partial y}\right)^2e^{2s}\sin^2 t\\ &\quad+\left(\frac{\partial u}{\partial x}\right)^2e^{2s}\sin^2 t-2\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}e^{2s}\sin t\cos t+\left(\frac{\partial u}{\partial y}\right)^2e^{2s}cos^2t=e^{2s}\left(\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2\right)\end{aligned}$$

例. 若 $z=f(x,y)$，$x=r\cos\theta$，$y=r\sin\theta$，證明 $\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2=\left(\frac{\partial z}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2$。

解. 由鏈鎖律

$$\begin{aligned}\frac{\partial z}{\partial r}&=\frac{\partial z}{\partial x}\frac{\partial x}{\partial r}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial r}=\frac{\partial z}{\partial x}\cos\theta+\frac{\partial z}{\partial y}\sin\theta\\ \frac{\partial z}{\partial \theta}&=\frac{\partial z}{\partial x}\frac{\partial x}{\partial \theta}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial \theta}=\frac{\partial z}{\partial x}\cdot(-r\sin\theta)+\frac{\partial z}{\partial y}\cdot(r\cos\theta)\end{aligned}$$

故

$$\begin{aligned}\left(\frac{\partial z}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2&=\left(\frac{\partial z}{\partial x}\cos\theta+\frac{\partial z}{\partial y}\sin\theta\right)^2+\frac{1}{r^2}\left(\frac{\partial z}{\partial x}\cdot(-r\sin\theta)+\frac{\partial z}{\partial y}\cdot(r\cos\theta)\right)^2\\ &=\left(\frac{\partial z}{\partial x}\right)^2\cos^2\theta+2\frac{\partial z}{\partial x}\frac{\partial z}{\partial y}\cos\theta\sin\theta+\left(\frac{\partial z}{\partial y}\right)^2\sin^2\theta\\ &\quad+\left(\frac{\partial z}{\partial x}\right)^2\sin^2\theta-2\frac{\partial z}{\partial x}\frac{\partial z}{\partial y}\sin\theta\cos\theta+\left(\frac{\partial z}{\partial y}\right)^2\cos^2\theta=\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2\end{aligned}$$

例. 若 $z=u(x,y)$，$x=r^2+s^2$，$y=2rs$，求 $\dfrac{\partial z}{\partial r}$，$\dfrac{\partial^2 z}{\partial r^2}$，$\dfrac{\partial^2 z}{\partial s\partial r}$。

解. 由鏈鎖律

$$
\begin{aligned}
\frac{\partial z}{\partial r} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial r}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial r}=2r\frac{\partial u}{\partial x}+2s\frac{\partial u}{\partial y}\\
\frac{\partial^2 z}{\partial r^2} &= \frac{\partial}{\partial r}\left(2r\frac{\partial u}{\partial x}+2s\frac{\partial u}{\partial y}\right)=2\frac{\partial u}{\partial x}+2r\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right)+2s\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial y}\right)\\
\frac{\partial^2 z}{\partial s\partial r} &= \frac{\partial}{\partial s}\left(2r\frac{\partial u}{\partial x}+2s\frac{\partial u}{\partial y}\right)=2\frac{\partial u}{\partial y}+2s\frac{\partial}{\partial s}\left(\frac{\partial u}{\partial y}\right)+2r\frac{\partial}{\partial s}\left(\frac{\partial u}{\partial x}\right)
\end{aligned}
$$

又

$$
\begin{aligned}
\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial r}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\frac{\partial y}{\partial r}=\frac{\partial^2 u}{\partial x^2}\cdot(2r)+\frac{\partial^2 u}{\partial y\partial x}\cdot(2s)\\
\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)\frac{\partial x}{\partial r}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial r}=\frac{\partial^2 u}{\partial x\partial y}\cdot(2r)+\frac{\partial^2 u}{\partial y^2}\cdot(2s)\\
\frac{\partial}{\partial s}\left(\frac{\partial u}{\partial x}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial s}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\frac{\partial y}{\partial s}=\frac{\partial^2 u}{\partial x^2}\cdot(2s)+\frac{\partial^2 u}{\partial y\partial x}\cdot(2r)\\
\frac{\partial}{\partial s}\left(\frac{\partial u}{\partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)\frac{\partial x}{\partial s}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial s}=\frac{\partial^2 u}{\partial x\partial y}\cdot(2s)+\frac{\partial^2 u}{\partial y^2}\cdot(2r)
\end{aligned}
$$

故

$$
\begin{aligned}
\frac{\partial^2 z}{\partial r^2} &= 2\frac{\partial u}{\partial x}+2r\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right)+2s\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial y}\right)\\
&= 2\frac{\partial u}{\partial x}+2r\left(\frac{\partial^2 u}{\partial x^2}\cdot(2r)+\frac{\partial^2 u}{\partial y\partial x}\cdot(2s)\right)+2s\left(\frac{\partial^2 u}{\partial x\partial y}\cdot(2r)+\frac{\partial^2 u}{\partial y^2}\cdot(2s)\right)\\
&= 2\frac{\partial u}{\partial x}+4r^2\frac{\partial^2 u}{\partial x^2}+8rs\frac{\partial^2 u}{\partial x\partial y}+4s^2\frac{\partial^2 u}{\partial y^2}\\
\frac{\partial^2 z}{\partial s\partial r} &= 2\frac{\partial u}{\partial y}+2s\frac{\partial}{\partial s}\left(\frac{\partial u}{\partial y}\right)+2r\frac{\partial}{\partial s}\left(\frac{\partial u}{\partial x}\right)\\
&= 2\frac{\partial u}{\partial y}+2s\left(\frac{\partial^2 u}{\partial x\partial y}\cdot(2s)+\frac{\partial^2 u}{\partial y^2}\cdot(2r)\right)+2r\left(\frac{\partial^2 u}{\partial x^2}\cdot(2s)+\frac{\partial^2 u}{\partial y\partial x}\cdot(2r)\right)\\
&= 2\frac{\partial u}{\partial y}+4rs\frac{\partial^2 u}{\partial x^2}+4(r^2+s^2)\frac{\partial^2 u}{\partial x\partial y}+4rs\frac{\partial^2 u}{\partial y^2}
\end{aligned}
$$

例. 若 $z=u(x,y)$，$x=g(s,t)$，$y=h(s,t)$，證明

$$
\frac{\partial^2 z}{\partial t^2}=\frac{\partial z}{\partial x}\frac{\partial^2 x}{\partial t^2}+\frac{\partial z}{\partial y}\frac{\partial^2 y}{\partial t^2}+\frac{\partial^2 z}{\partial x^2}\left(\frac{\partial x}{\partial t}\right)^2+2\frac{\partial^2 z}{\partial x\partial y}\frac{\partial x}{\partial t}\frac{\partial y}{\partial t}+\frac{\partial^2 z}{\partial y^2}\left(\frac{\partial y}{\partial t}\right)^2
$$

解. 令 $z=U(s,t)=u(x(s,t),\,y(s,t))$，由鏈鎖律

$$
\begin{aligned}
\frac{\partial U}{\partial t} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial t}\\
\frac{\partial^2 U}{\partial t^2} &= \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial t}\right)=\frac{\partial u}{\partial x}\frac{\partial^2 x}{\partial t^2}+\frac{\partial x}{\partial t}\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\right)+\frac{\partial u}{\partial y}\frac{\partial^2 y}{\partial t^2}+\frac{\partial y}{\partial t}\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial y}\right)
\end{aligned}
$$

又

$$\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial t}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\frac{\partial y}{\partial t}=\frac{\partial^2 u}{\partial x^2}\frac{\partial x}{\partial t}+\frac{\partial^2 u}{\partial y\partial x}\frac{\partial y}{\partial t}$$
$$\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial y}\right)=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)\frac{\partial x}{\partial t}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial t}=\frac{\partial^2 u}{\partial x\partial y}\frac{\partial x}{\partial t}+\frac{\partial^2 u}{\partial y^2}\frac{\partial y}{\partial t}$$

故

$$\begin{aligned}\frac{\partial^2 U}{\partial t^2}&=\frac{\partial u}{\partial x}\frac{\partial^2 x}{\partial t^2}+\frac{\partial x}{\partial t}\left(\frac{\partial^2 u}{\partial x^2}\frac{\partial x}{\partial t}+\frac{\partial u}{\partial y}\frac{\partial^2 y}{\partial t^2}+\frac{\partial^2 u}{\partial y\partial x}\frac{\partial y}{\partial t}\right)+\frac{\partial y}{\partial t}\left(\frac{\partial^2 u}{\partial x\partial y}\frac{\partial x}{\partial t}+\frac{\partial^2 u}{\partial y^2}\frac{\partial y}{\partial t}\right)\\&=\frac{\partial u}{\partial x}\frac{\partial^2 x}{\partial t^2}+\frac{\partial u}{\partial y}\frac{\partial^2 y}{\partial t^2}+\frac{\partial^2 u}{\partial x^2}\left(\frac{\partial x}{\partial t}\right)^2+2\frac{\partial^2 u}{\partial x\partial y}\frac{\partial x}{\partial t}\frac{\partial y}{\partial t}+\frac{\partial^2 u}{\partial y^2}\left(\frac{\partial y}{\partial t}\right)^2\end{aligned}$$

例. 若 $f(x,t)=g(x+at)+h(x-at)$，其中 g，h 可二次微分，證明 f 滿足波動方程式 $\dfrac{\partial^2 f}{\partial t^2}=a^2\dfrac{\partial^2 f}{\partial x^2}$。

解. 令 $u(x,t)=x+at$，$v(x,t)=x-at$，$f\big(u(x,t),v(x,t)\big)=g(u(x,t))+h(v(x,t))$。由鏈鎖律

$$\begin{aligned}\frac{\partial f}{\partial t}&=\frac{\partial f}{\partial u}\frac{\partial u}{\partial t}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial t}\\&=g'(u(x,t))\cdot a+h'(v(x,t))\cdot(-a)=a\,g'(x+at)-a\,h'(x-at)=a\,g'(u(x,t))-a\,h'(v(x,t))\\\frac{\partial^2 f}{\partial t^2}&=\frac{\partial}{\partial t}\left(\frac{\partial f}{\partial t}\right)\\&=\frac{\partial}{\partial u}\big(a\,g'(u)-a\,h'(v)\big)\big(u(x,t),v(x,t)\big)\frac{\partial u}{\partial t}+\frac{\partial}{\partial v}\big(a\,g'(u)-a\,h'(v)\big)\big(u(x,t),v(x,t)\big)\frac{\partial v}{\partial t}\\&=a\,g''(u(x,t))\cdot a-a\,h''(v(x,t))\cdot(-a)\\&=a^2\left(g''(x+at)+h''(x-at)\right)\\\frac{\partial f}{\partial x}&=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}\\&=g'(u(x,t))\cdot 1+h'(v(x,t))\cdot(1)=g'(x+at)+h'(x-at)=g'(u(x,t))+h'(v(x,t))\\\frac{\partial^2 f}{\partial x^2}&=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)\\&=\frac{\partial}{\partial x}\big(g'(u)+h'(v)\big)\big(u(x,t),v(x,t)\big)\frac{\partial u}{\partial x}+\frac{\partial}{\partial v}\big(g'(u)+h'(v)\big)\big(u(x,t),v(x,t)\big)\frac{\partial v}{\partial x}\\&=g''(u(x,t))\cdot 1+h''(v(x,t))\cdot 1\\&=g''(x+at)+h''(x-at)\end{aligned}$$

故 $\dfrac{\partial^2 f}{\partial t^2}=a^2\dfrac{\partial^2 f}{\partial x^2}$。

例. 若 $u=f(x,y)$，$x=e^s\cos t$，$y=e^s\sin t$，證明 $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=e^{-2s}\left(\dfrac{\partial^2 u}{\partial s^2}+\dfrac{\partial^2 u}{\partial t^2}\right)$。

解. 由鏈鎖律

$$\frac{\partial u}{\partial s}=\frac{\partial u}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial s}=\frac{\partial u}{\partial x}\cdot(e^s\cos t)+\frac{\partial u}{\partial y}\cdot(e^s\sin t)$$
$$\frac{\partial u}{\partial t}=\frac{\partial u}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial t}=\frac{\partial u}{\partial x}\cdot(-e^s\sin t)+\frac{\partial u}{\partial y}\cdot(e^s\cos t)$$

$$
\begin{aligned}
\frac{\partial^2 u}{\partial s^2} = \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial s}\right) &= \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial x}\cdot(e^s\cos t) + \frac{\partial u}{\partial y}\cdot(e^s\sin t)\right) \\
&= \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial x}\right)\cdot(e^s\cos t) + \frac{\partial u}{\partial x}\cdot(e^s\cos t) + \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial y}\right)\cdot(e^s\sin t) + \frac{\partial u}{\partial y}\cdot(e^s\sin t) \\
\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial t}\right) &= \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\cdot(-e^s\sin t) + \frac{\partial u}{\partial y}\cdot(e^s\cos t)\right) \\
&= \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\right)\cdot(-e^s\sin t) + \frac{\partial u}{\partial x}\cdot(-e^s\cos t) + \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial y}\right)\cdot(e^s\cos t) + \frac{\partial u}{\partial y}\cdot(-e^s\sin t)
\end{aligned}
$$

又

$$
\begin{aligned}
\frac{\partial}{\partial s}\left(\frac{\partial u}{\partial x}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial s} + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\frac{\partial y}{\partial s} = \frac{\partial^2 u}{\partial x^2}\cdot(e^s\cos t) + \frac{\partial^2 u}{\partial y\partial x}\cdot(e^s\sin t) \\
\frac{\partial}{\partial s}\left(\frac{\partial u}{\partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)\frac{\partial x}{\partial s} + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial s} = \frac{\partial^2 u}{\partial x\partial y}\cdot(e^s\cos t) + \frac{\partial^2 u}{\partial y^2}\cdot(e^s\sin t) \\
\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial t} + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\frac{\partial y}{\partial t} = \frac{\partial^2 u}{\partial x^2}\cdot(-e^s\sin t) + \frac{\partial^2 u}{\partial y\partial x}\cdot(e^s\cos t) \\
\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)\frac{\partial x}{\partial t} + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial t} = \frac{\partial^2 u}{\partial x\partial y}\cdot(-e^s\sin t) + \frac{\partial^2 u}{\partial y^2}\cdot(e^s\cos t)
\end{aligned}
$$

故

$$
\begin{aligned}
\frac{\partial^2 u}{\partial s^2} &= \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial x}\right)\cdot(e^s\cos t) + \frac{\partial u}{\partial x}\cdot(e^s\cos t) + \frac{\partial}{\partial s}\left(\frac{\partial u}{\partial y}\right)\cdot(e^s\sin t) + \frac{\partial u}{\partial y}\cdot(e^s\sin t) \\
&= \left(\frac{\partial^2 u}{\partial x^2}\cdot(e^s\cos t) + \frac{\partial^2 u}{\partial y\partial x}\cdot(e^s\sin t)\right)\cdot(e^s\cos t) + \frac{\partial u}{\partial x}\cdot(e^s\cos t) \\
&\quad + \left(\frac{\partial^2 u}{\partial x\partial y}\cdot(e^s\cos t) + \frac{\partial^2 u}{\partial y^2}\cdot(e^s\sin t)\right)\cdot(e^s\sin t) + \frac{\partial u}{\partial y}\cdot(e^s\sin t) \\
\frac{\partial^2 u}{\partial t^2} &= \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\right)\cdot(-e^s\sin t) + \frac{\partial u}{\partial x}\cdot(-e^s\cos t) + \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial y}\right)\cdot(e^s\cos t) + \frac{\partial u}{\partial y}\cdot(-e^s\sin t) \\
&= \left(\frac{\partial^2 u}{\partial x^2}\cdot(-e^s\sin t) + \frac{\partial^2 u}{\partial y\partial x}\cdot(e^s\cos t)\right)\cdot(-e^s\sin t) + \frac{\partial u}{\partial x}\cdot(-e^s\cos t) \\
&\quad + \left(\frac{\partial^2 u}{\partial x\partial y}\cdot(-e^s\sin t) + \frac{\partial^2 u}{\partial y^2}\cdot(e^s\cos t)\right)\cdot(e^s\cos t) + \frac{\partial u}{\partial y}\cdot(-e^s\sin t)
\end{aligned}
$$

可得 $\dfrac{\partial^2 u}{\partial s^2} + \dfrac{\partial^2 u}{\partial t^2} = e^{2s}\left(\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2}\right)$。

例. 若 $z = u(x,y)$，$x = r\cos\theta$，$y = r\sin\theta$，求 $\dfrac{\partial^2 z}{\partial\theta\partial r}$，$\dfrac{\partial^2 z}{\partial r\partial\theta}$。

解.

- 求 $\dfrac{\partial^2 z}{\partial\theta\partial r}$：由鏈鎖律

$$
\begin{aligned}
\frac{\partial z}{\partial r} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta \\
\frac{\partial^2 z}{\partial\theta\partial r} &= \frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta\right) \\
&= \frac{\partial u}{\partial x}(-\sin\theta) + \frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial x}\right)\cos\theta + \frac{\partial u}{\partial y}\cos\theta + \frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial y}\right)\sin\theta
\end{aligned}
$$

又

$$\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial\theta}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\frac{\partial y}{\partial\theta}=\frac{\partial^2 u}{\partial x^2}(-r\sin\theta)+\frac{\partial^2 u}{\partial y\partial x}(r\cos\theta)$$
$$\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial y}\right)=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)\frac{\partial x}{\partial\theta}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial\theta}=\frac{\partial^2 u}{\partial x\partial y}(-r\sin\theta)+\frac{\partial^2 u}{\partial y^2}(r\cos\theta)$$

故

$$\begin{aligned}\frac{\partial^2 z}{\partial\theta\partial r}&=\frac{\partial u}{\partial x}(-\sin\theta)+\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial x}\right)\cos\theta+\frac{\partial u}{\partial y}\cos\theta+\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial y}\right)\sin\theta\\&=\frac{\partial u}{\partial x}(-\sin\theta)+\left(\frac{\partial^2 u}{\partial x^2}(-r\sin\theta)+\frac{\partial^2 u}{\partial y\partial x}(r\cos\theta)\right)\cos\theta\\&\quad+\frac{\partial u}{\partial y}\cos\theta+\left(\frac{\partial^2 u}{\partial x\partial y}(-r\sin\theta)+\frac{\partial^2 u}{\partial y^2}(r\cos\theta)\right)\sin\theta\\&=\frac{\partial u}{\partial y}\cos\theta-\frac{\partial u}{\partial x}\sin\theta+r\sin\theta\cos\theta\left(\frac{\partial^2 u}{\partial y^2}-\frac{\partial^2 u}{\partial x^2}\right)+r(\cos^2\theta-\sin^2\theta)\frac{\partial^2 u}{\partial x\partial y}\end{aligned}$$

- 求 $\dfrac{\partial^2 z}{\partial r\partial\theta}$：由鏈鎖律

$$\begin{aligned}\frac{\partial z}{\partial\theta}&=\frac{\partial u}{\partial x}\frac{\partial x}{\partial\theta}+\frac{\partial u}{\partial y}\frac{\partial y}{\partial\theta}=\frac{\partial u}{\partial x}(-r\sin\theta)+\frac{\partial u}{\partial y}(r\cos\theta)\\\frac{\partial^2 z}{\partial r\partial\theta}&=\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}(-r\sin\theta)+\frac{\partial u}{\partial y}(r\cos\theta)\right)\\&=\frac{\partial u}{\partial x}(-\sin\theta)+\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right)(-r\sin\theta)+\frac{\partial u}{\partial y}\cos\theta+\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial y}\right)(r\cos\theta)\end{aligned}$$

又

$$\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial r}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\frac{\partial y}{\partial r}=\frac{\partial^2 u}{\partial x^2}\cos\theta+\frac{\partial^2 u}{\partial y\partial x}\sin\theta$$
$$\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial y}\right)=\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)\frac{\partial x}{\partial r}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial r}=\frac{\partial^2 u}{\partial x\partial y}\cos\theta+\frac{\partial^2 u}{\partial y^2}\sin\theta$$

故

$$\begin{aligned}\frac{\partial^2 z}{\partial r\partial\theta}&=\frac{\partial u}{\partial x}(-\sin\theta)+\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right)(-r\sin\theta)+\frac{\partial u}{\partial y}\cos\theta+\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial y}\right)(r\cos\theta)\\&=\frac{\partial u}{\partial x}(-\sin\theta)+\left(\frac{\partial^2 u}{\partial x^2}\cos\theta+\frac{\partial^2 u}{\partial y\partial x}\sin\theta\right)(-r\sin\theta)\\&\quad+\frac{\partial u}{\partial y}\cos\theta+\left(\frac{\partial^2 u}{\partial x\partial y}\cos\theta+\frac{\partial^2 u}{\partial y^2}\sin\theta\right)(r\cos\theta)\\&=\frac{\partial u}{\partial y}\cos\theta-\frac{\partial u}{\partial x}\sin\theta+r\sin\theta\cos\theta\left(\frac{\partial^2 u}{\partial y^2}-\frac{\partial^2 u}{\partial x^2}\right)+r(\cos^2\theta-\sin^2\theta)\frac{\partial^2 u}{\partial x\partial y}\end{aligned}$$

例. 若 $z=u(x,y)$，$x=r\cos\theta$，$y=r\sin\theta$，證明 $\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=\dfrac{\partial^2 z}{\partial r^2}+\dfrac{1}{r}\dfrac{\partial z}{\partial r}+\dfrac{1}{r^2}\dfrac{\partial^2 z}{\partial\theta^2}$。

解. 令 $z = U(r,\theta) = u(x(r,\theta),\, y(r,\theta))$，由鏈鎖律

$$
\begin{aligned}
\frac{\partial U}{\partial r} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta \\
\frac{\partial^2 U}{\partial r^2} &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta\right)\frac{\partial x}{\partial r} + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta\right)\frac{\partial y}{\partial r} \\
&= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta\right)\cos\theta + \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta\right)\sin\theta \\
&= \frac{\partial^2 u}{\partial x^2}\cos^2\theta + 2\frac{\partial^2 u}{\partial x\partial y}\sin\theta\cos\theta + \frac{\partial^2 u}{\partial y^2}\sin^2\theta \\
\frac{\partial U}{\partial \theta} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta} = \frac{\partial u}{\partial x}(-r\sin\theta) + \frac{\partial u}{\partial y}(r\cos\theta) \\
\frac{\partial^2 U}{\partial \theta^2} &= \frac{\partial}{\partial \theta}\left(\frac{\partial u}{\partial x}(-r\sin\theta) + \frac{\partial u}{\partial y}(r\cos\theta)\right) \\
&= \frac{\partial u}{\partial x}(-r\cos\theta) - r\sin\theta\,\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial x}\right) + \frac{\partial u}{\partial y}(-r\sin\theta) + r\cos\theta\,\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial y}\right) \\
&= \frac{\partial u}{\partial x}(-r\cos\theta) + \frac{\partial u}{\partial y}(-r\sin\theta) - r\sin\theta\,\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial x}\right) + r\cos\theta\,\frac{\partial}{\partial\theta}\left(\frac{\partial u}{\partial y}\right) \\
&= \frac{\partial u}{\partial x}(-r\cos\theta) + \frac{\partial u}{\partial y}(-r\sin\theta) - r\sin\theta\left(\frac{\partial^2 u}{\partial x^2}\frac{\partial x}{\partial\theta} + \frac{\partial^2 u}{\partial y\partial x}\frac{\partial y}{\partial\theta}\right) + r\cos\theta\left(\frac{\partial^2 u}{\partial x\partial y}\frac{\partial x}{\partial\theta} + \frac{\partial^2 u}{\partial y^2}\frac{\partial y}{\partial\theta}\right) \\
&= \frac{\partial u}{\partial x}(-r\cos\theta) + \frac{\partial u}{\partial y}(-r\sin\theta) - r\sin\theta\left(\frac{\partial^2 u}{\partial x^2}(-r\sin\theta) + \frac{\partial^2 u}{\partial y\partial x}(r\cos\theta)\right) \\
&\qquad + r\cos\theta\left(\frac{\partial^2 u}{\partial x\partial y}(-r\sin\theta) + \frac{\partial^2 u}{\partial y^2}(r\cos\theta)\right) \\
&= \frac{\partial u}{\partial x}(-r\cos\theta) + \frac{\partial u}{\partial y}(-r\sin\theta) + r^2\sin^2\theta\,\frac{\partial^2 u}{\partial x^2} - 2r^2\sin\theta\cos\theta\,\frac{\partial^2 u}{\partial y\partial x} + r^2\cos^2\theta\,\frac{\partial^2 u}{\partial y^2}
\end{aligned}
$$

由上，

$$
\begin{aligned}
\frac{\partial^2 U}{\partial r^2} + \frac{1}{r}\frac{\partial U}{\partial r} + \frac{1}{r^2}\frac{\partial^2 U}{\partial\theta^2} &= \left(\frac{\partial^2 u}{\partial x^2}\cos^2\theta + 2\frac{\partial^2 u}{\partial x\partial y}\sin\theta\cos\theta + \frac{\partial^2 u}{\partial y^2}\sin^2\theta\right) + \frac{1}{r}\left(\frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta\right) \\
&\quad + \frac{1}{r^2}\left(\frac{\partial u}{\partial x}(-r\cos\theta) + \frac{\partial u}{\partial y}(-r\sin\theta) + r^2\sin^2\theta\,\frac{\partial^2 u}{\partial x^2} - 2r^2\sin\theta\cos\theta\,\frac{\partial^2 u}{\partial y\partial x} + r^2\cos^2\theta\,\frac{\partial^2 u}{\partial y^2}\right) \\
&= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}
\end{aligned}
$$

5.5 方向導數與梯度

定義. 令 $S \subseteq \mathbb{R}^n$，$\mathbf{c} \in S$，$\mathbf{u} \in \mathbb{R}^n$，可微函數 $f : S \to \mathbb{R}$；$\mathbb{R}^n$ 中直角座標系單位向量為 $\mathbf{e}_1,\, \mathbf{e}_2,\, \ldots,\, \mathbf{e}_n$。

- f 在 $\mathbf{c}$ 之 $\mathbf{u}$ 方向導數 (directional derivative) 為 $D_{\mathbf{u}}f(\mathbf{c}) = \lim\limits_{h\to 0}\dfrac{f(\mathbf{c}+h\mathbf{u}) - f(\mathbf{c})}{h}$。
- f 之偏導數 $f_i(\mathbf{c}) = D_{\mathbf{e}_i}f(\mathbf{c})$，$i = 1,\, 2,\, \ldots,\, n$。
- f 在 $\mathbf{c}$ 之梯度 (gradient) 為 $\nabla f(\mathbf{c}) = \langle f_1(\mathbf{c}),\, f_2(\mathbf{c}),\, \ldots,\, f_n(\mathbf{c})\rangle$。

性質. $D_{\mathbf{u}}f(\mathbf{c}) = \nabla f(\mathbf{c}) \cdot \mathbf{u}$。

證. 令 $g(x) = f(\mathbf{c}+x\mathbf{u}) = f(v_1+xu_1,\, v_2+xu_2,\, \ldots,\, v_n+xu_n)$,$D_{\mathbf{u}}f(\mathbf{c}) = \lim\limits_{h\to 0}\dfrac{f(\mathbf{c}+h\mathbf{u}) - f(\mathbf{c})}{h} = \lim\limits_{h\to 0}\dfrac{g(h) - g(0)}{h} = g'(x)\big|_{x=0}$。由鏈鎖律，$g'(x)\big|_{x=0} = \displaystyle\sum_{i=1}^{n} f_i(\mathbf{c}+x\mathbf{u})\,\frac{\mathrm{d}(v_i+xu_i)}{\mathrm{d}x}\bigg|_{x=0} = \sum_{i=1}^{n} f_i(\mathbf{c}+x\mathbf{u})\,u_i\bigg|_{x=0} = \nabla f(\mathbf{c}) \cdot \mathbf{u}$。

性質. 給定 $\mathbb{R}^n$ 曲面 $G(\mathbf{x})=0$。令 $\mathbf{x}_0\in\mathbb{R}^n$ 使 $G(\mathbf{x}_0)=0$ ($\mathbf{x}_0$ 位於 $G(\mathbf{x})$ 上)，則 $\nabla G(\mathbf{x}_0)$ 在 $\mathbf{x}_0$ 與 $G(\mathbf{x})$ 垂直。

證. 令 $\mathbb{R}^n$ 曲線 $\mathbf{r}(t)=\langle x_1(t),\, x_2(t),\, \ldots,\, x_n(t)\rangle$ 位於 $G(\mathbf{x})$ 上且通過 $\mathbf{x}_0$，則 $G(\mathbf{r}(t))=0$，且存在 $t_0\in\mathbb{R}$ 使 $\mathbf{r}(t_0)=\mathbf{x}_0$。由鏈鎖律，$G(\mathbf{r}(t))=0$ 等式兩邊對 t 微分並代入 $t=t_0$ 得

$$\left.\frac{\mathrm{d}G(\mathbf{r}(t))}{\mathrm{d}t}\right|_{t=t_0}=0 \implies \left.\sum_{i=1}^{n} G_i(\mathbf{r}(t))\, x_i'(t)\right|_{t=t_0}=0 \implies \nabla G(\mathbf{r}(t_0))\cdot\mathbf{r}'(t_0)=0$$

例. 求 $z=x^2+5xy-2y^2$ 在 $(1,2,3)$ 之切平面方程式。

解. $f(x,y,z)=x^2+5xy-2y^2-z=0$，故 $\nabla f=(2x+5y)\widehat{\boldsymbol{\imath}}+(-4y+5x)\widehat{\boldsymbol{\jmath}}-\widehat{\mathbf{k}}$，$\nabla f(1,2,3)=\langle 12,\, -3,\, -1\rangle$，切平面方程式為 $12(x-1)-3(y-2)-(z-3)=0 \implies 12x-3y-z=3$。

5.6 極值問題

定義. 給定 $R\subseteq\mathbb{R}^n$，$f:R\to\mathbb{R}$，$B(\mathbf{x},h)\equiv\{\mathbf{y}\in\mathbb{R}^n \mid |\mathbf{y}-\mathbf{x}|<h\}$。

- f 在 $\mathbf{x}_{\mathrm{M}}\in R$ 有最大值 (global maximum) $f(\mathbf{x}_{\mathrm{M}})$：$f(\mathbf{x}_{\mathrm{M}})\geqslant f(\mathbf{x}),\ \forall\,\mathbf{x}\in R$。
- f 在 $\mathbf{x}_{\mathrm{m}}\in R$ 有最小值 (global minimum) $f(\mathbf{x}_{\mathrm{m}})$：$f(\mathbf{x}_{\mathrm{m}})\leqslant f(\mathbf{x}),\ \forall\,\mathbf{x}\in R$。
- f 在 $\mathbf{x}_0\in R$ 有極大值 (local maximum) $f(\mathbf{x}_0)$：$\exists\, h_0>0$ 使 $f(\mathbf{x}_0)\geqslant f(\mathbf{x}),\ \forall\,\mathbf{x}\in B(\mathbf{x}_0,h_0)\cap R$。
- f 在 $\mathbf{x}_1\in R$ 有極小值 (local minimum) $f(\mathbf{x}_1)$：$\exists\, h_1>0$ 使 $f(\mathbf{x}_1)\leqslant f(\mathbf{x}),\ \forall\,\mathbf{x}\in B(\mathbf{x}_1,h_1)\cap R$。

5.7 Lagrange 乘數法

$$\begin{aligned} &\min_{x}\ f(x) \\ \text{subject to}\quad & g_i(x)=0,\quad i=1,2,\ldots,l. \end{aligned}$$